

Project Charter

1. Project Title

Library Borrowing Data Analysis for Hsinchu City Library

2. Project Purpose

The Hsinchu City Library has collected extensive borrowing records from its readers.

The purpose of this project is to analyze borrowing patterns to:

- Understand popular book categories and seasonal trends.
- Identify differences in reading behavior between age groups.
- Provide recommendations to improve library services and book acquisitions.

3. Goals & Objectives

- Build a dashboard that visualizes borrowing trends by category, time, and user group.
- Perform frequent itemset mining (Apriori algorithm) to discover commonly borrowed book combinations.
- Conduct RFM segmentation to classify active and inactive readers.
- Provide actionable insights for library management to optimize resource allocation.

4. Project Scope

In Scope:

- Use the open dataset Library Borrowing Records from the government open data portal.
- Apply data cleaning, frequent itemset mining, RFM analysis, and visualization methods.
- Deliver interactive dashboard and analysis report.

Out of Scope:

- Integration with the library live IT system.
- Building a recommendation system for real-time user interactions (future phase).

5. Key Tasks & Deliverables

- Data Collection & Cleaning → Import open data CSV, handle missing values, reformat timestamps.
- Exploratory Data Analysis (EDA) → Identify seasonal borrowing patterns and user demographics.
- Frequent Itemset Mining → Apply Apriori algorithm to find common borrowing patterns.
- RFM Segmentation → Segment readers by Recency, Frequency, and Monetary.
- Deliverables: EDA report, itemset mining results, RFM segmentation table, dashboard visualizations.

6. Roles & Responsibilities

- Project Sponsor: Library Director (approves project, defines strategic needs).
- Project Manager: Faculty Advisor (guides students, ensures progress).
- Data Scientist(s): Perform cleaning, analysis, and modeling.
- Data Engineer: Support data pipeline preparation.
- Student Team Members: Contribute to analysis, coding, and visualization.

7. Timeline / Milestones

Phase	Timeline (Weeks)	Deliverable
Data Collection & Cleaning	Week 1–2	Cleaned dataset
EDA & Visualization	Week 3–4	EDA report with charts
Itemset Mining & RFM	Week 5–6	Analysis results
Final Report & Dashboard	Week 7–8	Dashboard & final presentation

8. Approval & Authorization

By signing this document, the sponsor authorizes the project team to begin work according to the scope and objectives defined.

- Project Sponsor: _____

- Project Manager: _____

- Date: _____